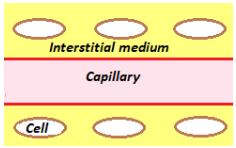
19 BIOPHYSICS OF SOLUTIONS *Pr B. Boutabia-Chéraitia*

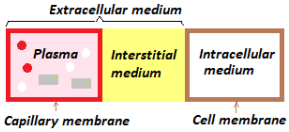
- Water and solutes are transported across **membranes** (cellular and capillary). These membranes are classified into three types:

- **Dialysis**: permeable to **water** and **small molecules** or **ions** $(Na^+, K^+, Cl^-, \dots)$, and impermeable to **macromolecules** $(M \geqslant 10kg)$.



- **Semi-permeable**: permeable only to **water**.

- **Selective** (or **biological**): permeable to **water** and **certain molecules**.



G- COLLIGATIVES PROPERTIES

Physico-chemical properties of solutions that depend only on the **number of solute particles** that constitute them.

1 - EBULLIOSCOPIC ELEVATION

$$Solution = Solvent + Solute$$

20 BIOPHYSICS OF SOLUTIONS *Pr B. Boutabia-Chéraitia*

We designate by

T_0 : vaporization temperature of the **solvent**.

T_s : vaporization temperature of the **solution**.

$$T_s > T_0 \Rightarrow \Delta T_{vap} = T_s - T_0 > 0$$

ΔT_{vap} is called **ebullioscopic elevation** and is given by **Raoult**'s law:

$$\Delta T_{vap} = K_{eb} \times \omega \qquad [\Delta T_{vap}] = °C$$

ω : osmolarity of the solution. $[\omega] = osmol/l$

K_{eb} : ebullioscopic constante of the solvent. $[K_{eb}] = °C.l/osmol$

Remarks

If ω : osmolality $\Rightarrow [\omega] = osmol/kg \Rightarrow [K_{eb}] = °C.kg/osmol$

If $[\Delta T] = K \Rightarrow [K_{eb}] = K.l/osmol$, or, $[K_{eb}] = K.kg/osmol$

2 - CRYOSCOPIC DEPRESSION

T_0 : solidification -or freezing- temperature of the **solvent**.

21 BIOPHYSICS OF SOLUTIONS *Pr B. Boutabia-Chéraitia*

T_s : solidification temperature of the **solution**.

$$T_s < T_0 \Rightarrow \Delta T_{sol} = T_s - T_0 < 0$$

ΔT_{sol} is called **cryoscopic depression** and is given by **Raoult**'s law:

$$\Delta T_{sol} = -K_{cr} \times \omega \qquad [\Delta T_{sol}] = °C$$

ω : osmolarity of the solution. $[\omega] = osmol/l$

K_{cr} : cryoscopic constant of the solvent. $[K_{cr}] = °C.l/osmol$

Remarks

If ω : osmolality $\Rightarrow [\omega] = osmol/kg \Rightarrow [K_{cr}] = °C.kg/osmol$

If $[\Delta T] = K \Rightarrow [K_{cr}] = K.l/osmol$, or, $[K_{cr}] = K.kg/osmol$

Exercise 6:

The cryoscopic depression of an alcohol-intoxicated subject is $-0.62°C$. What would his blood alcohol concentration (g/l) be if all other blood parameters are normal?

22 BIOPHYSICS OF SOLUTIONS *Pr B. Boutabia-Chéraitia*

Given: Normal cryoscopic depression of blood: $-0.56°C; M_{alc} = 46g/mol$; for plasma water: $K_{cr} = 1.86°C.l/osmol$

Solutions:

Normal blood $(\Delta T_1, \omega_1) \rightarrow \Delta T_1 = -K_{cr} \times \omega_1$

Blood containing alcohol $(\Delta T_2, \omega_2) \rightarrow \Delta T_2 = -K_{cr} \times \omega_2$

$$\omega_2 = \omega_1 + \omega_{alc} \Rightarrow \Delta T_2 = -K_{cr}(\omega_1 + \omega_{alc}) = -K_{cr} \times \omega_1 - K_{cr} \times \omega_{alc}$$

$$\Rightarrow \Delta T_2 = \Delta T_1 - K_{cr} \times \omega_{alc}$$

$$\Rightarrow \omega_{alc} = \frac{\Delta T_2 - \Delta T_1}{-K_{cr}} = \frac{-0.62-(-0.56)}{-1.86} = 0.032 osmol/l$$

Ethanol: non electrolyte $\Rightarrow C_{alc} = \omega_{alc}$

$\Rightarrow$ Blood alcohol concentration:

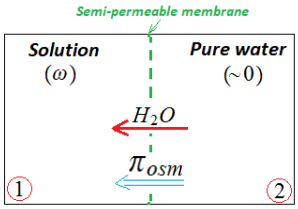
$$C_p = C \times M = 0.032 \times 46 = 1.48 g/l$$

3 - OSMOTIC PRESSURE

The diffusion of water through a **semi-permeable** membrane is called **osmosis**.

23 BIOPHYSICS OF SOLUTIONS *Pr B. Boutabia-Chéraitia*

$$H_2O : (2) \rightarrow (1)$$



This diffusion occurs under the effect of a **force** called **osmotic pressure** (π_{osm}).

$$\pi_{osm} : (2) \rightarrow (1) \qquad [\pi_{osm}] = Pa$$

$$\pi_{osm} = \omega RT \quad \textbf{\textit{Van't Hoff}'s law.}$$

π_{osm} : measures the tendency of a solution to **draw in** water by **osmosis**.

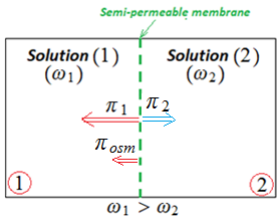
In a general way

(1) draw in H_2O with $\pi_1 = \omega_1 RT$.

(2) draw in H_2O with $\pi_2 = \omega_2 RT$.

$\omega_1 > \omega_2 \Rightarrow \Delta\pi \equiv \pi_{osm} = (\omega_1 - \omega_2)RT$

$H_2O : (2) \rightarrow (1)$ until the osmolarities are equal.



24 BIOPHYSICS OF SOLUTIONS *Pr B. Boutabia-Chéraitia*

Exercise 7:

At $25°C$, two compartments (1) and (2) separated by a perfectly semi-permeable membrane contain, respectively, an aqueous solution of $NaCl$ at $5.85g/l$, and an aqueous solution of glucose at $18g/l$.
In which direction does the water flow? And under what pressure?

Solutions:

$$\omega_1 = 2C_1 = 2\frac{5,85}{58.5} = 0.2osmol/l = 0.2 \times 10^3 osmol/m^3$$

$$\omega_2 = C_2 = \frac{18}{180} = 0.1osmol/l = 0.1 \times 10^3 osmol/m^3$$

$\omega_1 > \omega_2 \Rightarrow$ water diffuses from $(2) \rightarrow (1)$ under the effect of osmotic pressure, $\pi_{osm} : (2) \rightarrow (1)$.

$$\pi_{osm} = (\omega_1 - \omega_2)RT = 0.1 \times 10^3 \times 8.31 \times 298 = 248kPa$$

Remarks

⇝ **Non-diffusible** particles induce water movement and thereby generate **osmotic pressure**. They are termed **osmotically active**.

25 BIOPHYSICS OF SOLUTIONS *Pr B. Boutabia-Chéraitia*

$\rightsquigarrow$ If $\pi_{osm} = 0 \Leftrightarrow \pi_1 = \pi_2 \Leftrightarrow$ no water flow $\Rightarrow (1)$ and (2) are said to be **isotonic** or **iso-osmotic**.

$\rightsquigarrow$ If $\pi_1 < \pi_2 \Rightarrow (1)$: **hypo-tonic** and (2) : **hyper-tonic**.

$\rightsquigarrow$ **Isotonicity** $(\pi_{osm} = 0)$ and **semi-permeable** membrane,

$\pi_1 = \pi_2 \Rightarrow \omega_1 = \omega_2$: **iso-osmolarity**.

$\rightsquigarrow$ **Isotonicity** $(\pi_{osm} = 0)$ and **biologic** membrane $\Rightarrow$ **not necessarily** iso-osmolarity.

Example

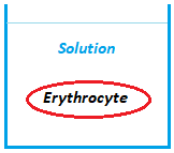
An erythrocyte $(300mosmol/l)$ is immersed in a solution containing $300mosmol/l$ of $NaCl$ and $100mosmol/l$ of urea, which diffuses freely across its membrane.

Erythrocyte: (1)

$$\omega_1 = 300mosmol/l = 300osmol/m^3$$
$$\pi_1 = \omega_1 RT = 300RT$$

Solution: (2)

$$\omega_2 = \omega_{NaCl} + \omega_{urea} = 400mosmol/l$$



26 BIOPHYSICS OF SOLUTIONS *Pr B. Boutabia-Chéraitia*

Urea can be treated as water (pure solvent) because it diffuses across the erythrocyte membrane $\Rightarrow$ it is **not osmotically active**.

$\Rightarrow \pi_2 = \omega_{NaCl} RT = 300RT$

We have iso-tonicity $(\pi_1 = \pi_2)$ but no iso-osmolarity $(\omega_1 \neq \omega_2)$.

Remarks

$\rightsquigarrow$ if the membrane is **totally permeable** to the solute) $\Rightarrow \pi_{osm} = 0$

$\rightsquigarrow$ If the membrane is **partially permeable** to the solute $\Rightarrow \pi_{osm} = \sigma\omega RT$

σ : Solute **reflection** coefficient

$$\sigma = \frac{Number\ of\ osmoles\ that\ do\ not\ cross\ the\ membrane}{Initial\ number\ of\ osmoles} = \frac{Area\ of\ the\ pores\ impermeable\ to\ the\ solute}{Total\ pore\ area}$$

$\rightsquigarrow$ If the membrane is **dialysis** $\Rightarrow \sigma = 0$

$\rightsquigarrow$ If the membrane is **semi-permeable** $\Rightarrow \sigma = 1$

$0 \leqslant \sigma \leqslant 1$

$\rightsquigarrow$ If there are i solutes $(\sigma_i, \omega_i) \Rightarrow \pi_{osm} = \sum_i \sigma_i \omega_i RT$